

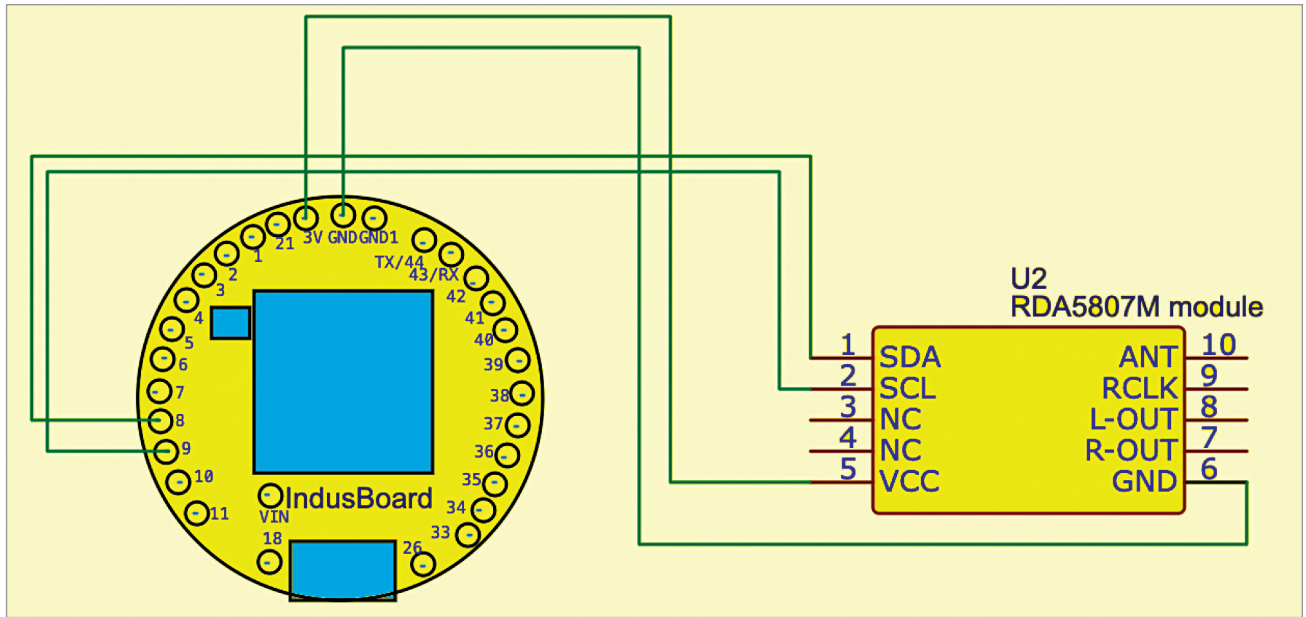


Fig. 2: Circuit for programming the board to set the desired FM frequency and adjust the volume

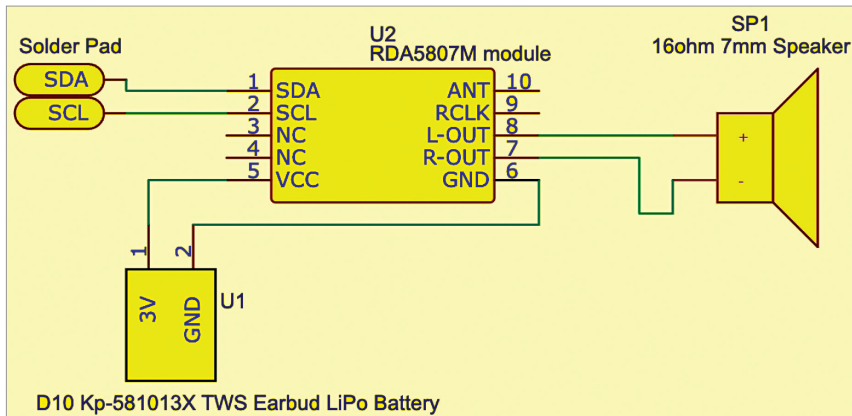


Fig. 3: FM earbud circuit for AirPods

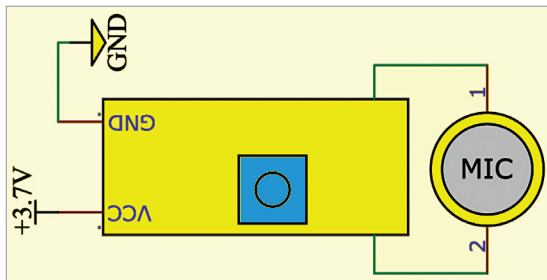


Fig. 4: FM radio transmitter circuit

Software

To configure the FM earbud, begin by setting the FM IC to the desired frequency and adjusting the volume.

Volume, bass, and other settings can also be adjusted by sending the appropriate commands. To view all

Connect the IC to the MCU using the SDA and SCL pins, as shown in Figs. 2 through 4. Then, install the radio library in the Arduino IDE and upload the code.

To set the radio station, send the command `Fnnnnn` (where `nnnnn` is the desired

frequency) via the serial interface.

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available command functions, send ? Help via the serial interface. Fig. 5 shows a snippet of the source code.

Construction

Refer to Fig. 6 before installing the battery, FM IC, and TWS earbud speaker inside the AirPods enclosure. Before configuring the frequency and volume, connect the FM radio chip to the TWS earbud speaker and battery as shown in the circuits in Figs. 2, 3, and 4.

Once assembled, the FM AirPod-style earbud is ready for use, enabling reception of radio signals on the configured frequency. For use as a spy device, connect the FM audio transmitter as illustrated in Fig. 4.